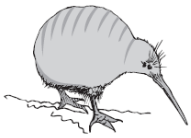
MRS GREN

Classification is a word used to describe the grouping of things according to their similarities. One of the most basic types of classification is to decide whether something is living or non-living. Living things are called organisms.


a fly


a camp fire


kiwi


clouds


cabbage

The five things above can be readily separated into living and non-living. Living – fly, kiwi, cabbage. Non-living – fire, clouds.

Here are the seven activities of all living things. If you remember that someone has an auntie called MRS GREN you will be able to remember all seven activities.

- Movement
- Respiration
- Sensitivity
- Growth
- Reproduction
- Excretion
- Nutrition

Most of this is common sense. You can move around from place to place and so can most animals. A few animals like barnacles can only do this when they are young larvae. Adult barnacles are firmly stuck onto the rock, but they can still move their legs to catch food. Animals move to: find shelter, escape from predators and other danger, find food, and find a mate.

Respiration is the release of energy from glucose or another organic chemical in our cells. Respiration happens in ALL living things.

Sensitivity is very important. How would you know if something was fit to eat if you could not smell or taste it? How would you escape danger? Plants sense water – their roots grow towards it.

Growth. We all do it.

Reproduction: Continuation of the species.

Excretion means getting rid of the poisonous waste products from the chemical reactions in our bodies. We breathe out carbon dioxide and we urinate.

Nutrition. We have to eat to provide the materials for growth, repair and to provide us with energy.

Why was fire classified as non-living?

To decide if something is living or non-living, run through each of the seven criteria of MRS GREN. If the answer is no to any of them, then the object is non-living.

For fire:

- Movement – yes
- Respiration – no (uses oxygen and releases carbon dioxide, but obviously is not occurring in cells)
- Sensitivity – yes/no (can respond to wind/water, etc)
- Growth – yes
- Reproduction – yes (a spark from one fire will cause another fire to start)
- Excretion – yes (the smoke)
- Nutrition – yes (the wood, paper, etc that it burns)

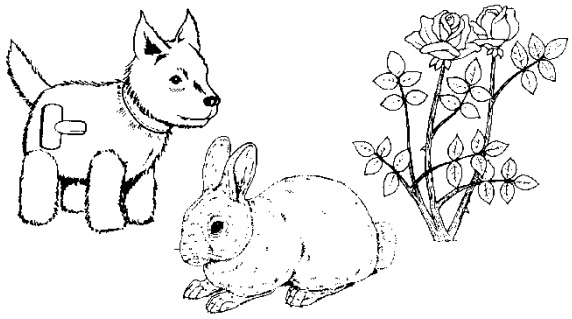
As the answer was ‘no’ to respiration, fire is non-living, even though it carries out all the other activities. Respiration and breathing are not the same.

Breathing is a mechanical process. When we breathe in, our diaphragm contracts and pulls down the base of our lungs, and intercostal muscles move the ribs out, stretching and enlarging the lungs, so that air is sucked in down our windpipe (via our nose or mouth). As we breathe out, our diaphragm and chest muscles relax, the lungs are no longer stretched and air is pushed back out of the lungs. Respiration involves the transfer of energy. The respiratory system is made up of the lungs and airways such as the windpipe, throat and nasal passages. Our lungs absorb oxygen from the air we breathe. The oxygen is transported around our body and used to 'burn up' the food we eat and convert it to energy. This results in the production of carbon dioxide, which is breathed out.

Excretion is not getting rid of solid waste (faeces) but the elimination of the waste products of the chemical processes inside the cells (e.g. in urine, in exhaled air).

Questions

1. What is another word that could be used to describe 'classification'?
2. Split the following things up into living and non-living: a mushroom, a table, moss, butter, bacteria, ferns, distilled water, the wind, a cat.
3. Why were clouds classified as non-living? Run through each of the seven criteria of MRS GREN like the example for "fire" on the previous page.
4. The drawings show a wind-up toy dog, a rabbit and a rose bush.



Use **only** the words in the list below to complete **all** the sentences. You may use the words more than once.

eat grow move from place to place reproduce

(i) In what way are the toy dog and the rabbit similar? Both the toy dog and the rabbit can

(ii) Give **two** ways in which the toy dog is different from the rabbit. (a) The toy dog cannot…(b) The toy dog cannot….

(iii) Give **two** ways in which the rabbit is different from the rose bush. The rabbit can…

5. Copy and complete this table. Remember even plants move their leaves to get sunlight.

	APPLE TREE	TIGER	TOY ROBOT	FOSSIL	COMP-UTER	SNAKE	AERO-PLANE
Does it Move?	✓						
Does it Respire?	✓						
Does it have Sensitivity?	✓						
Does it Grow?							
Does it Reproduce?							
Does it Excrete?							
Does it need Nutrition?							
SO IS IT ALIVE?	Yes						

6. In your own words, describe the difference between breathing and respiration.